Software Requirements Specification (SRS)

AI-Driven Commercial Vehicle Safety and Violation Detection System

| Offline-first edge system for accident prevention via vision, GIS and driver-behavior monitoring

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1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for an embedded, offline-first vehicle safety system that detects and predicts unsafe driving conditions — human error, incorrect driving technique, vehicle overload, and poor driver behavior — and intervenes in real time through in-cabin alerts and fleet-level reporting.

1.2 Scope

The system monitors a commercial vehicle's interior, exterior, and movement using onboard cameras and sensors; analyzes this data locally using AI; correlates vehicle speed against upcoming road geometry (curves, gradients); and raises alerts before a violation becomes an accident. It operates fully without network connectivity and opportunistically syncs to a fleet dashboard when connectivity is available. It does not control the vehicle (no braking/throttle intervention) — it is an advisory and logging system, not an autonomous-driving system.

1.3 Definitions & Abbreviations

Term	Meaning
Edge device	On-vehicle compute unit (Raspberry Pi 5 / Jetson Orin Nano) running all inference locally
GNSS/GPS	Satellite positioning receiver
GIS	Geographic Information System (road geometry, elevation data)
Blackbox	Local tamper-resistant event/evidence recorder
Debouncing	Technique to prevent false alerts from momentary/noisy detections
DPDP Act	Digital Personal Data Protection Act, 2023 (India)

1.4 References

Institute patent "System for Monitoring Active Safety of a Vehicle and Method Thereof" (conceptual reference only — all algorithms in this SRS are independently specified); MoRTH Road Accidents in India reports; IRC:38 (horizontal curve design); DPDP Act 2023.

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2. Overall Description

2.1 Product Perspective

A self-contained, battery-powered, retrofit-free unit installed per vehicle. It is not wired into existing
vehicle electronics (prevents tampering, simplifies installation across heterogeneous fleets). It must be fully operational with zero connectivity; the fleet dashboard is a secondary, best-effort layer.

2.2 Product Functions (summary)

- Detect driver fatigue, distraction, and phone use
- Count and monitor cabin occupancy against capacity
- Detect upcoming road curves/gradients and compute a safe approach speed
- Monitor vehicle load (Phase 2) and overtaking safety (Phase 2)
- Arbitrate detections into confirmed violations, filtering sensor noise
- Alert the driver in-cabin; log evidence to a local Blackbox
- Sync to a fleet dashboard when network is available
- Produce a per-driver safety score over time

2.3 User Classes

User	Access
Driver (primary user)	In-cabin visual/audio alerts only; no access to recorded footage
Fleet Operator (secondary user)	Dashboard: violation summaries, compliance reports, vehicle map
Transport Authority (secondary user)	Evidence-backed violation logs for regulatory review
System Admin	Device configuration, user/role management, data retention settings

2.4 Operating Environment

Embedded Linux on edge hardware; intermittent cellular connectivity; outdoor-range temperature tolerance; mountainous/hilly terrain with GNSS dropouts (tunnels, valleys, dense cover).

2.5 Constraints

- Must function fully offline — no requirement may depend on live connectivity to execute core safety logic
- Low-cost, commodity hardware only (no proprietary vehicle bus access)
- No invasive vehicle modification
- Must comply with DPDP Act 2023 for any personal/biometric-adjacent data (driver face, voice if applicable)
- Real-time constraint: alert latency must stay within a safety-relevant window (target: <500 ms from detection to in-cabin alert).

2.6 Assumptions & Dependencies

Vehicle has a stable 12V power source for charging the unit's battery; cameras have an unobstructed view; local OSM/SRTM data for the operating region is pre-loaded before deployment.

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3. System Features (Functional Requirements)

Grouped by delivery phase, per your feature-priority table.

PHASE 1 : Core / MVP

FR-1 Exterior (Forward) Camera Monitoring Continuously capture forward road view; detect vehicles/obstacles ahead; record rolling buffer for evidence clips.

FR-2 Interior (Driver) Camera Monitoring Capture driver's face/upper body; feed to driver-behavior analysis (FR-8); processed in-memory, never stored as raw video.

FR-3 Vehicle & Movement Parameter Sensing Acquire GPS position, speed, heading, and IMU (accelerometer/gyroscope) data at a fixed sampling interval (e.g., 1–5 Hz).

FR-4 AI Analysis Engine Local inference pipeline running vision models (face/landmark, object detection, tracking) and the GIS/speed-logic module, entirely on-device.

FR-5 Risk Detection Classify incoming sensor/vision outputs into risk categories: drowsiness, distraction, phone use, overspeed-for-curve, overcrowding, (later: overload, unsafe overtake).

FR-6 Safe-Speed Prediction — CORE Given current curve radius, gradient, and vehicle speed, compute a recommended safe approach speed; flag when actual speed exceeds it beyond a defined margin.

FR-7 Curve Detection & Anticipation — CORE Map-match GPS position to the road network; identify the next curve within a look-ahead distance; retrieve curvature radius and turn proximity from locally cached GIS data (must work with GPS loss — see NFR-1).

FR-8 Driver Behavior Monitoring Detect eyelid closure duration, yawning frequency, head pose (distraction), and mobile phone handling; classify into drowsy/distracted/attentive states.

FR-9 Passenger Counting Detect and track persons in the cabin; compare live count against configured permitted capacity; flag overcrowding.

FR-10 Alert & Notification System Trigger in-cabin visual/audio alerts for confirmed violations (see FR-13 arbitration); alert severity levels (advisory vs. critical).

FR-11 Blackbox (Local Event Recorder) Record every confirmed violation event with timestamp, GPS location, sensor snapshot, and a short evidence clip; store encrypted; tamper-resistant local storage independent of main vehicle systems.

FR-12 Database & Data Management Structured local storage (events, driver profile, vehicle profile) with scheduled opportunistic sync to the central database when connectivity exists.

FR-13 Safety Quantification Compute a rolling per-driver safety score from violation frequency/severity over a configurable time window; expose to fleet dashboard.

FR-13a Violation Arbitration (supporting FR-10/FR-11) Before any alert fires: require multi-frame confirmation and confidence-weighted debouncing; cross-check against a second signal where available; suppress alerts dependent on a degraded/missing sensor.

PHASE 2 : Near-Term Extensions

FR-14 Load / Weight Monitoring Pressure sensors at wheels/luggage compartments/rooftop carrier; compute total load and distribution; flag unsafe imbalance or overload.

FR-15 Overtaking Analysis Using forward camera + relative speed/positioning, classify an overtake as safe/unsafe based on sight distance, oncoming traffic presence, and road width.

PHASE 3 : Advanced

FR-16 Expert-Driver Route Learning Model Pre-train a model on footage from an expert driver traversing a given route, capturing curve-by-curve "correct" driver response; use this as an additional, location-independent benchmark that works even where GIS data is unavailable.

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4. External Interface Requirements

4.1 Hardware

3× camera units (driver-facing, cabin, forward) — USB/CSI; GNSS/GPS receiver; IMU (accelerometer + gyroscope); optional pressure sensors (Phase 2); edge compute module (Raspberry Pi 5 / NVIDIA
Jetson Orin Nano); dedicated battery; in-cabin alert output (buzzer/LED/small display).

4.2 Software

Embedded Linux; Python-based inference runtime; local relational/embedded database; containerized services for maintainability.

4.3 Communication

MQTT (opportunistic, low-bandwidth telemetry) + REST/HTTPS (bulk sync, evidence clip upload) over cellular, only when available — no requirement in Section 3 may depend on this interface being up.

4.4 User Interface

In-cabin: minimal, non-distracting visual + audio alert only (no screen requiring driver attention while in motion). Fleet dashboard: web app (map view, violation list, driver scorecards, reports).

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5. Non-Functional Requirements

NFR-1 Offline-First Operation (critical) All Phase-1 functional requirements must execute with zero network connectivity. GIS data (OSM + SRTM) for the operating region is pre-cached locally. GPS loss (tunnels/valleys) is handled via IMU dead-reckoning + cached-tile map matching for a bounded grace window, entering a conservative "reduced-confidence" mode rather than failing.

NFR-2 Performance End-to-end detection-to-alert latency target: <500 ms. Minimum sustained inference rate: 10–15 FPS per camera stream on target edge hardware.

NFR-3 Reliability System must recover automatically from a power cycle or camera/sensor dropout without data loss of already-logged events; Blackbox storage must survive a crash of the main
inference process.

NFR-4 Security & Privacy Raw driver/cabin video never leaves the device and is discarded post-inference; only event metadata and short, violation-triggered clips persist, encrypted at rest (AES-256); role-based access on the dashboard; compliant with DPDP Act 2023 (consent, data minimization,
defined retention window, e.g. 30–90 days).

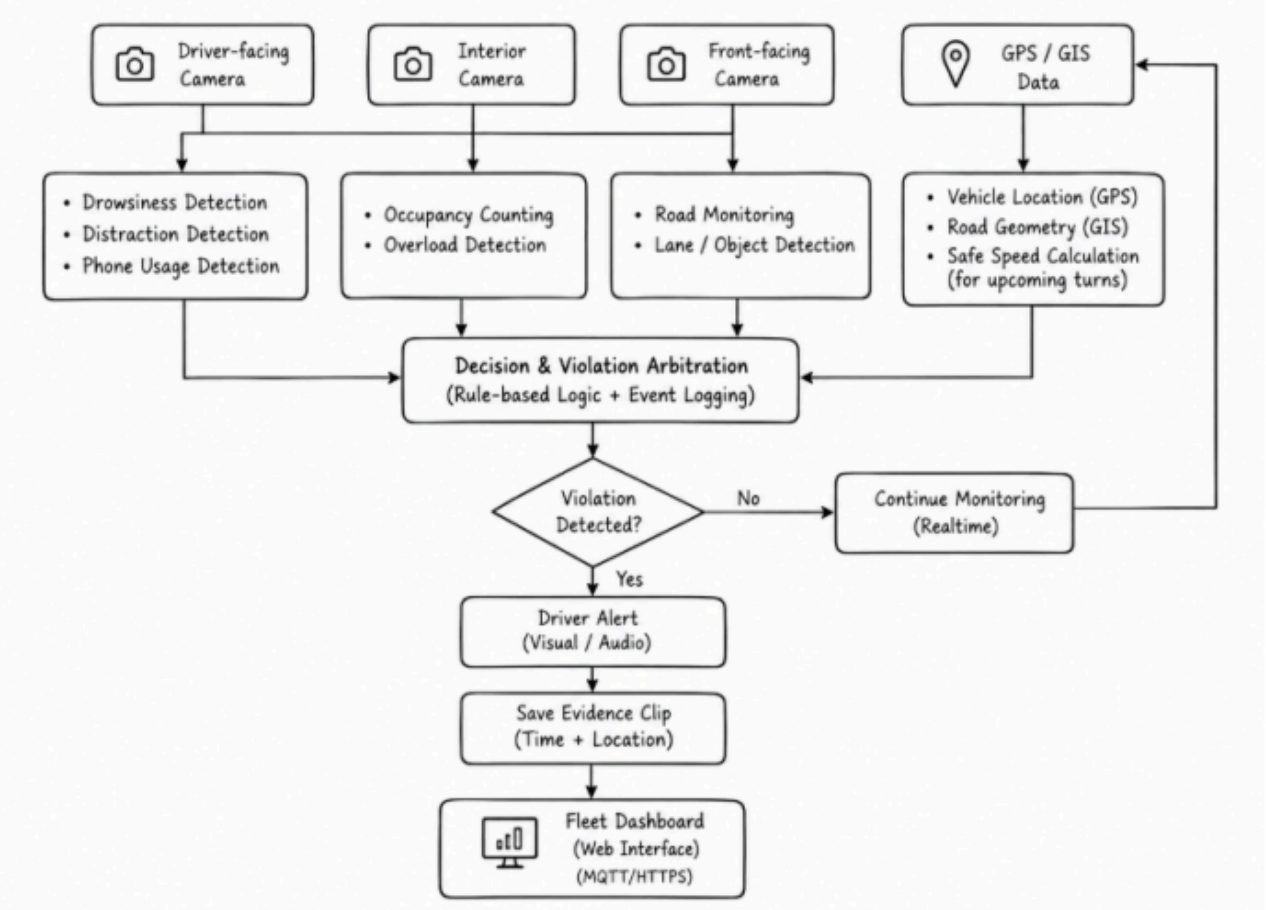
NFR-5 Usability In-cabin alerts must be interpretable within 1–2 seconds without requiring the driver to divert visual attention from the road.

NFR-6 Maintainability Modular architecture (each detection capability as an independent service) so models/thresholds can be updated without redeploying the whole system.

NFR-7 Scalability Backend/dashboard must support fleet sizes from a handful of vehicles to hundreds without redesign (horizontal scaling of ingestion and storage).

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6. System Architecture (high level)



7. Technology Stack

Category	Technologies / Tools
Hardware	Edge AI Module (Raspberry Pi 5 / NVIDIA Jetson Orin Nano), USB/CSI Camera Units (×3), GNSS/GPS Receiver, IMU (accelerometer/gyroscope), optional pressure sensors (Phase 2)

Programming Languages	Python (inference, backend), JavaScript (dashboard)
Vision & ML Frameworks	OpenCV, MediaPipe (Face Mesh), Ultralytics YOLO, ByteTrack, ONNX Runtime / TensorRT (edge acceleration)
GIS & Spatial Database	PostgreSQL with PostGIS, OpenStreetMap data, SRTM elevation data, Valhalla (map matching)
Local/Edge Storage	SQLite / SQLCipher (encrypted Blackbox storage)
Backend Framework	FastAPI
Messaging / Sync	MQTT (Mosquitto broker), REST/HTTPS
Visualization & Dashboard	React.js, Leaflet.js, Chart.js
Security	AES-256 at-rest encryption, role-based access control, DPDP Act 2023–aligned data handling
Deployment & Ops	Docker (backend/dashboard services), pytest (testing)

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8. Data Requirements (summary)

Entity	Key Fields
Vehicle	id, registration, capacity, assigned driver
Driver	id, name, license, rolling safety score
Violation Event	id, vehicle_id, type, timestamp, GPS location, confidence, evidence_clip_ref
Evidence Clip	id, encrypted_path, retention_expiry
GIS Cache	road_segment_id, curvature_radius, gradient, geometry

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9. Traceability Matrix (Patent Feature → Requirement → Phase)

#	**Patent Feature**	**SRS Requirement(s)**	**Phase**
1	Exterior camera	FR-1	1
2	Interior camera	FR-2	1
3	Vehicle/movement parameters	FR-3	1
4	AI analysis	FR-4	1

#	Patent Feature	SRS Requirement(s)	Phase
5	Risk detection	FR-5	1
6	Safe-speed prediction	FR-6	1 (CORE)
7	Curve detection/anticipation	FR-7	1 (CORE)
8	Driver safety monitoring	FR-8	1
9	Passenger counting	FR-9	1
10	Load monitoring	FR-14	2
11	Overtaking analysis	FR-15	2
12	Alerts	FR-10	1
13	Blackbox	FR-11	1
14	Database	FR-12	1
15	Safety quantification	FR-13	1
16	Expert-driver route model	FR-16	3 (Advanced)

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10. Application Modules

The system exposes three distinct user-facing modules, each scoped to a different user class from Section 2.3 and a different subset of functional requirements. All three consume the same backend/database (FR-12) but enforce strict role-based access — no module can see data outside its Scope.

10.1 Driver Module

User class: Primary user (driver) | Form factor: In-cabin hardware unit (primary) + optional companion mobile app (off-duty review)

Feature	Description	Linked FR
Login / trip start-end	Driver authenticates at shift start; trip is logged against their ID	FR-3, FR-12
Real-time alert display	Visual + audio alert for confirmed violations only (post-arbitration)	FR-10, FR-13a
Personal safety score	View own rolling safety score and trend over time	FR-13
Own trip/violation history	List of past violations on their trips — metadata only, **no cabin footage access**	FR-11, FR-12
Device health status	Simple indicator if a camera/sensor is obstructed or offline	FR-1–FR-3
Manual SOS (optional)	Driver-triggered emergency alert to fleet operator	FR-10

10.2 Passenger Module

User class: New — passengers of monitored vehicles (relevant to the public-transit use case) | Form factor: Mobile/web app

Feature	Description	Linked FR
Vehicle/route safety rating	Public-facing summary rating for a bus/route, derived from aggregated, anonymized safety scores (not per-driver detail)	FR-13
Live occupancy status	Seats-available / at-capacity indicator before boarding	FR-9
Trip tracking	Live vehicle position and ETA on the assigned route	FR-3
SOS / panic alert	Passenger-triggered alert routed to fleet operator and, if critical, transport authority	FR-10
Feedback / concern report	Free-text report tied to a specific trip, visible to fleet operator	— (new, logging-only)

10.3 Government / Admin Module

User classes: Fleet Operator + Transport Authority (two permission tiers within one module) | Form factor: Web dashboard (React/Leaflet/Chart.js, per Section 7)

Feature	Description	Linked FR	Tier
Live fleet map	Real-time vehicle positions, Leaflet-based	FR-3	Both
Violation log + filters	Browse/filter by vehicle, driver, date, violation type	FR-11, FR-12	Both
Evidence clip access	View time/location-stamped clips for a specific confirmed violation — every access is itself logged (audit trail)	FR-11	Both, access logged
Driver safety scorecards	Per-driver and fleet-wide safety score trends	FR-13	Both
Compliance report export	PDF/CSV export for regulatory submission	FR-12	Transport Authority
Vehicle/driver/user management	CRUD for vehicles, driver profiles, dashboard user accounts	— (admin)	Fleet Operator (own fleet only)
Passenger feedback inbox	Review concern reports submitted via the Passenger Module	—	Fleet Operator
Cross-fleet audit view	Aggregate, anonymized compliance view across multiple operators in a region	FR-13	Transport Authority only

Role separation note: a Fleet Operator account is scoped to its own vehicles only; a Transport Authority account can view evidence-backed logs and aggregate compliance data across all registered fleets in its jurisdiction, but still cannot access a driver's or passenger's data outside a

logged,
violation-triggered context — consistent with NFR-4.

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11. Hardware Devices: Existing Infrastructure vs. New Installation

11.1 Existing JKRTC Infrastructure (as publicly documented, early–mid 2026)

Per official statements from the Transport Minister, J&K Government, JKRTC's current fleet (buses and trucks) is already fitted with GPS tracking devices, with staff vehicles being the only exception [ref: J&K govt statement, Feb 2026]. Separately, a newer and much smaller batch — roughly 200 electric buses being inducted under the PM e-Drive scheme — is being equipped with a fuller Intelligent Transport Management System: real-time passenger information displays, panic buttons, automatic vehicle
location, CCTV cameras, a public address system, and stop-request buttons, all linked to the Integrated Command and Control Centres of Jammu/Srinagar Smart City Limited [ref: official rollout statement
reported by The Kashmir Walla, 2026].

Gap this project fills: the above is surveillance and location-tracking infrastructure, not AI-based safety inference. Neither the GPS-only legacy fleet nor the newer CCTV-equipped e-buses currently perform driver drowsiness/distraction detection, curve-aware speed advisory, on-device violation arbitration, or encrypted local evidence logging. This system is additive, not duplicative:

- On the ~200 newer e-buses: the existing GPS feed (and possibly the existing CCTV, subject to integration access) can potentially be reused; only the AI-analysis edge module, driver-facing camera, and IMU would need to be added.
- On the larger legacy fleet (GPS-only, no cameras): the full hardware kit below is required.

11.2 New Hardware To Be Installed (per vehicle)

Device	Purpose	Linked FR	Approx. Cost (INR)*
Driver-facing camera (IR/low-light capable)	Drowsiness, distraction, phone-use detection	FR-2, FR-8	₹1,500 – 3,000
Cabin-facing camera	Occupant counting	FR-9	₹800 – 2,000
Forward-facing dashcam	Obstacle/vehicle detection, road evidence	FR-1	₹800 – 2,500
GNSS/GPS receiver module	Position, speed (reusable on newer e-buses)	FR-3	₹500 – 1,500
IMU (accelerometer + gyroscope)	Dead-reckoning fallback in GPS-denied zones	FR-3, NFR-1	₹200 – 800
Edge compute — Raspberry Pi 5 (8GB)	On-device AI inference (budget option)	FR-4	₹6,000 – 9,000

or NVIDIA Jetson Orin Nano	On-device AI inference (GPU-accelerated option)	FR-4	₹20,000 – 30,000
Industrial-grade microSD/storage (128GB+)	Local OS + Blackbox storage	FR-11	₹1,500 – 2,500
Self-contained battery/power unit	Independent of vehicle electronics (tamper resistance)	Constraint §2.5	₹2,000 – 4,000
In-cabin alert unit (buzzer/LED or small display)	Driver alert	FR-10	₹300 – 1,000
4G cellular modem/SIM	Opportunistic dashboard sync	§4.3	₹1,500 – 3,000
Enclosure + mounting hardware	Installation, weatherproofing	—	₹1,000 – 2,000